

all-sites-exploratory

Linda Herrera

Packages

```
library(tidyverse)
library(here)
library(janitor)
library(viridis)
library(wesanderson)
```

Data

```
compiled_isotope <- read_csv(here("data", "compiled.csv"), show_col_types =
FALSE) |>
  clean_names() |> # lowercase and underscores in column names
  rename(sample = sample_name, # renaming columns to be shorter
         delta_d = processed_delta_d,
         delta_d_std = processed_delta_d_stddev,
         delta_o = processed_delta_18o,
         delta_o_std = processed_delta_18o_stddev) |>
  mutate(date = mdy(date)) # telling r the date format

gradient_tc <- read_csv(here("data", "tc_gradient_data.csv"))

gradient_cc <- read_csv(here("data", "cc_gradient_data.csv"))

metadata <- read_csv(here("data", "metadata.csv"))

trail_creek <- compiled_isotope |>
  filter(location %in% c("utc", "tc"))

coal_creek <- compiled_isotope |>
  filter(location %in% c("cc")) |>
  filter(site != "coal_3")

coal_creek_all <- compiled_isotope |>
  filter(location %in% c("cc"))
```

```
italian_creek <- compiled_isotope |>
  filter(location == "ic")
```

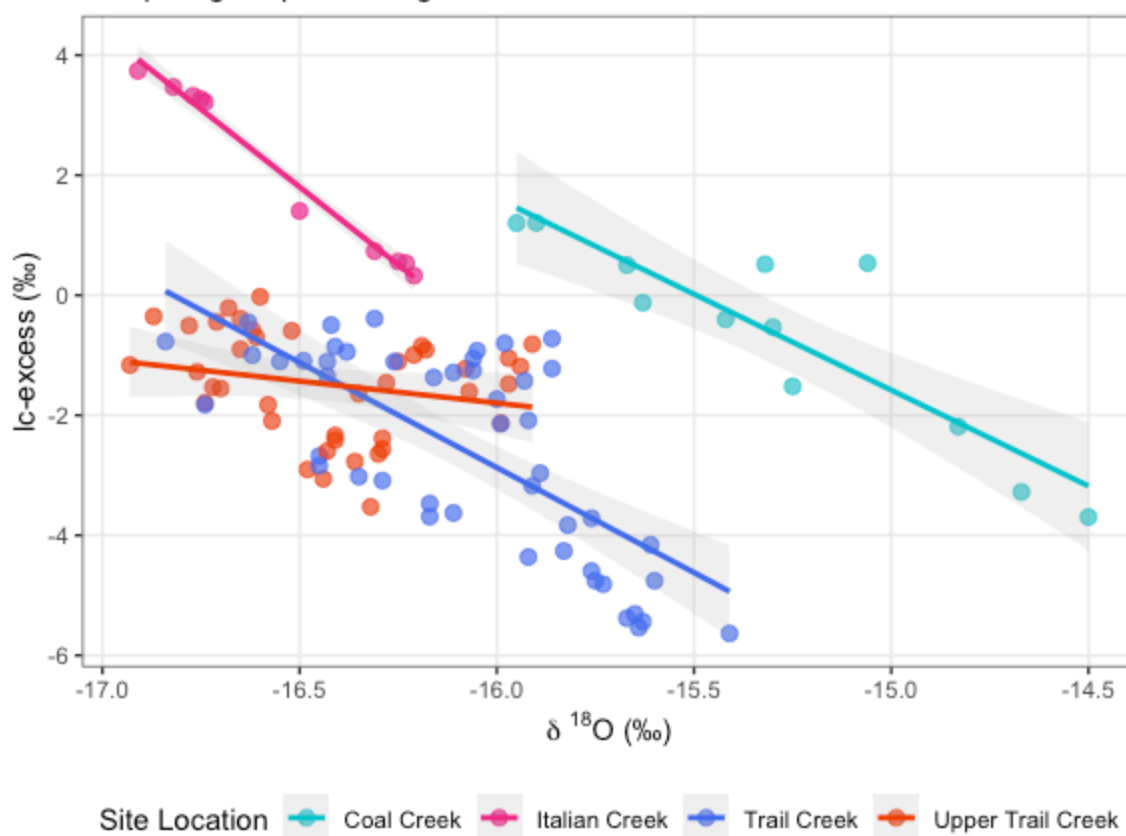
Exploratory Visuals with ALL Sites

Line-Conditioned Excess (lc-excess) vs Delta 18O

```
ggplot(data = compiled_isotope, aes(x = delta_o, y = lc_excess, color = location))
+
  geom_point(size = 2.5, alpha = 0.7) +
  geom_smooth(method = "lm", se = TRUE, alpha = 0.15, linewidth = 1) +
  scale_color_manual(
    values = c("tc" = "royalblue2", "cc" = "turquoise3", "it" = "deeppink2",
"utc" = "orangered2"),
    labels = c("cc" = "Coal Creek", "it" = "Italian Creek", "tc" = "Trail Creek",
"utc" = "Upper Trail Creek")
  ) +
  labs(
    title = "Line-Conditioned Excess (lc-excess) vs. Delta 180",
    subtitle = "Comparing evaporation signals between sites",
    x = expression(paste(delta, " ^18, "0 (\u2030)")), # Formats as  $\delta^{18}O$  (‰)
    y = "lc-excess (\u2030)",
    color = "Site Location") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
    legend.position = "bottom",
    panel.grid.minor = element_blank())
```

Line-Conditioned Excess (lc-excess) vs. Delta 18O

Comparing evaporation signals between sites



Time-Series of Line-Conditioned Excess (lc-excess)

```
ggplot(data = compiled_isotope,
  aes(x = date,
    y = lc_excess,
    color = location,
    group = location)) +
  geom_hline(yintercept = 0,
    linetype = "dashed",
    color = "gray50",
    size = 0.8) +
  geom_point(size = 2.5,
    alpha = 0.8) +
  geom_line(linewidth = 1,
    alpha = 0.7) +
  labs(title = "Time-Series of Line-Conditioned Excess (lc-excess)",
    subtitle = "Values below 0 indicate potential evaporative enrichment",
```

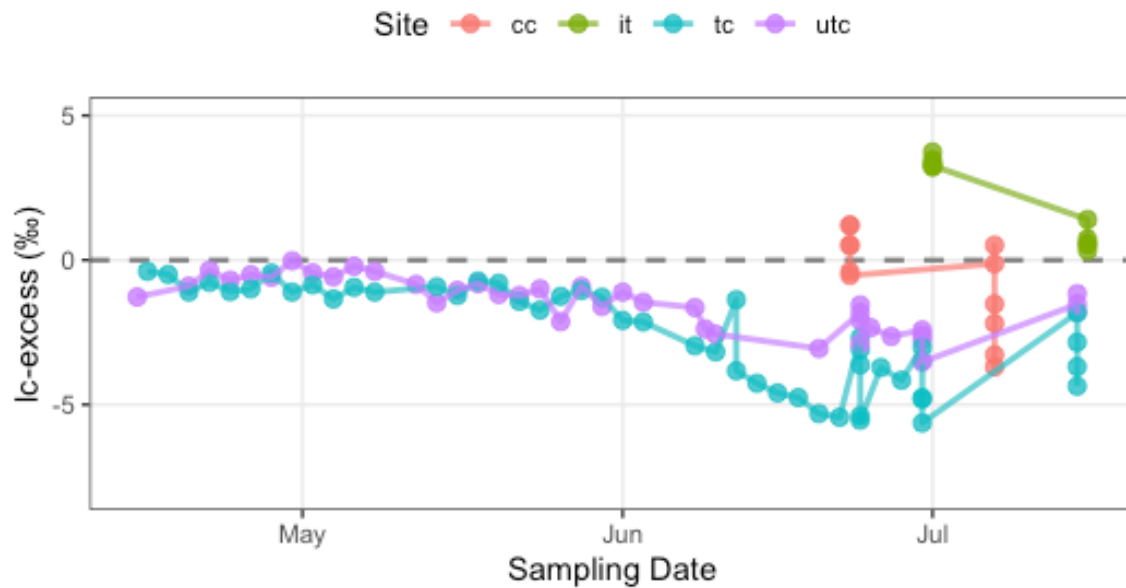
```

    x = "Sampling Date",
    y = "lc-excess (\u2030)",
    color = "Site") +
  ylim(-8,5) +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        panel.grid.minor = element_blank(),
        legend.position = "top")

```

Time-Series of Line-Conditioned Excess (lc-excess)

Values below 0 indicate potential evaporative enrichment



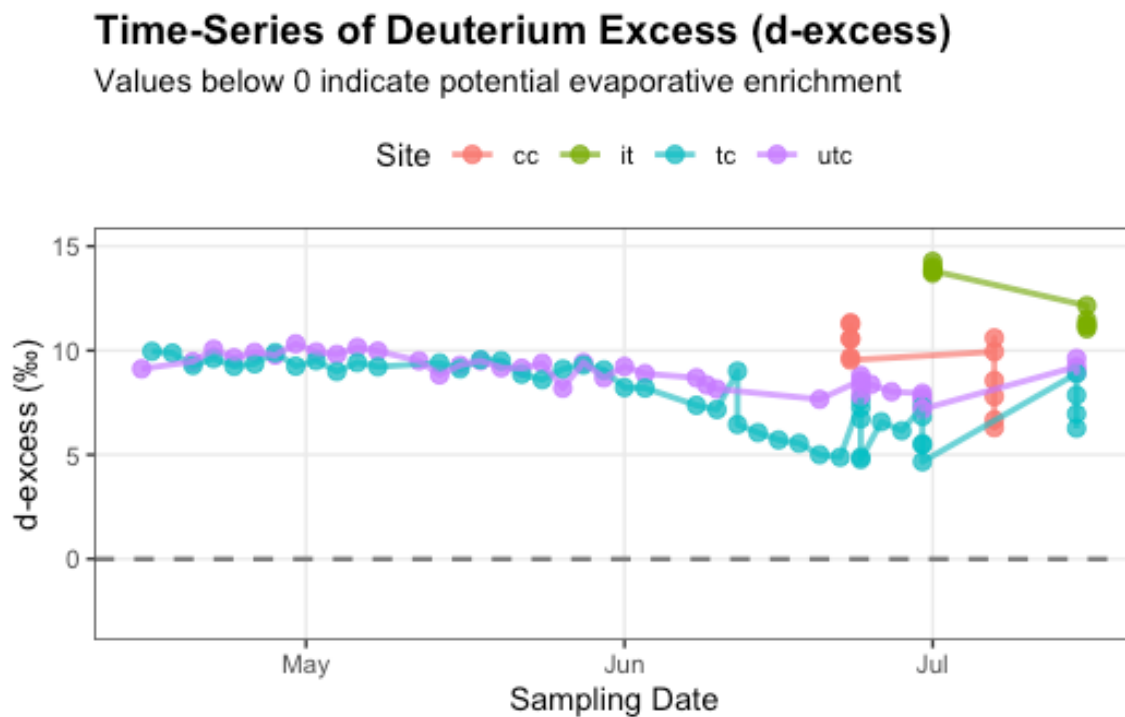
Time-Series of Deuterium Excess (d-excess)

```

ggplot(data = compiled_isotope,
  aes(x = date,
    y = d_excess,
    color = location,
    group = location)) +
  geom_hline(yintercept = 0,
    linetype = "dashed",
    color = "gray50",
    size = 0.8) +
  geom_point(size = 2.5,
    alpha = 0.8) +

```

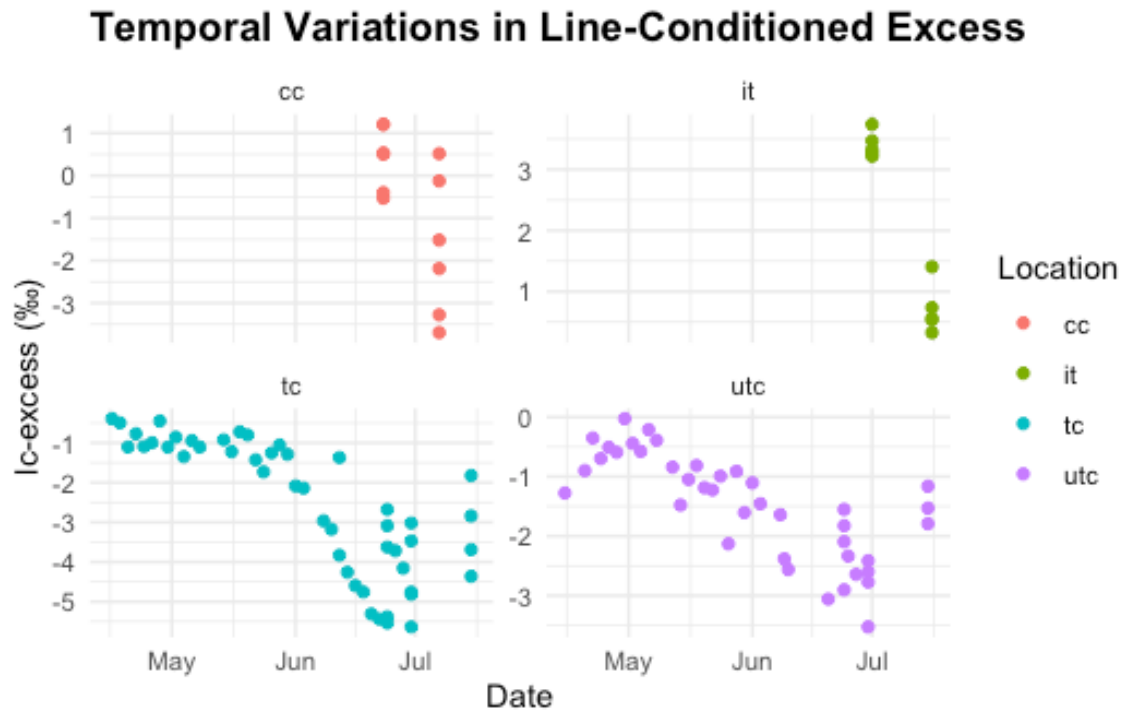
```
geom_line(size = 1,
          alpha = 0.7) +
labs(title = "Time-Series of Deuterium Excess (d-excess)",
     subtitle = "Values below 0 indicate potential evaporative enrichment",
     x = "Sampling Date",
     y = "d-excess (‰)",
     color = "Site") +
ylim(-3,15) +
theme_bw() +
theme(plot.title = element_text(face = "bold", size = 14),
      panel.grid.minor = element_blank(),
      legend.position = "top")
```



Temporal Variations in Line-Conditioned Excess (by location)

```
ggplot(compiled_isotope, aes(x = date, y = lc_excess, color = location)) +
geom_point(size = 1.5) +
labs(
  title = "Temporal Variations in Line-Conditioned Excess",
  x = "Date",
  y = "lc-excess (‰)",
  color = "Location")
```

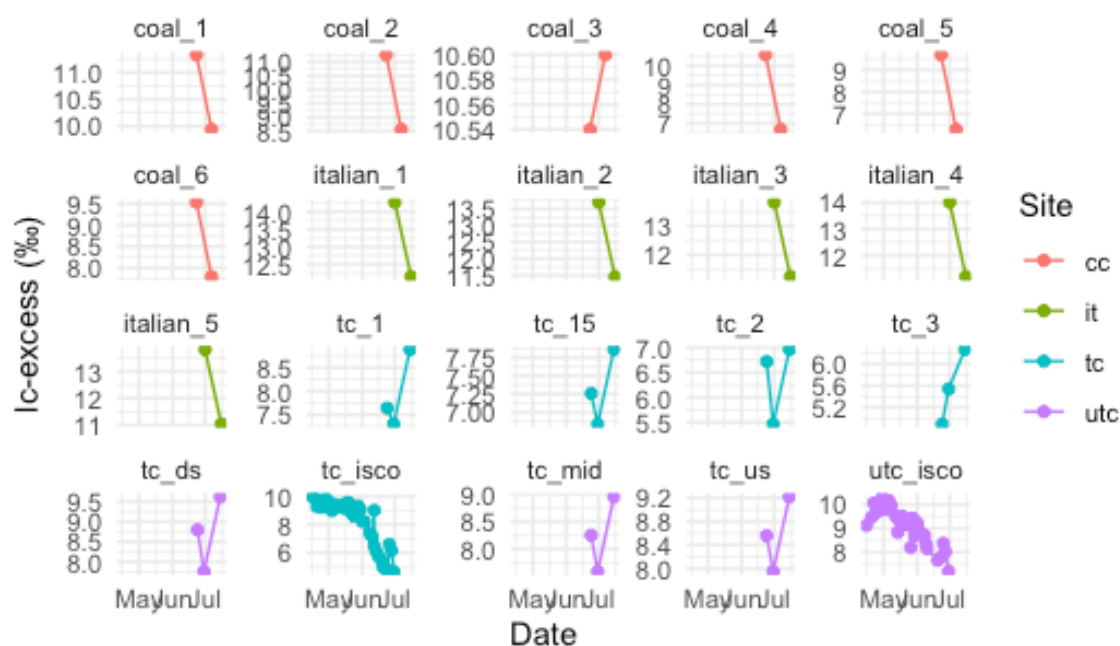
```
) +
facet_wrap(~location, scales = "free_y") + # Compare across sites easily
theme_minimal() +
theme(plot.title = element_text(face = "bold", size = 14))
```



Temporal Variations in Deuterium Excess (by sites)

```
ggplot(compiled_isotope, aes(x = date, y = d_excess, color = location)) +
  geom_line(linewidth = 0.5) +
  geom_point(size = 1.5) +
  labs(
    title = "Temporal Variations in Deuterium Excess",
    x = "Date",
    y = "lc-excess (‰)",
    color = "Site"
  ) +
facet_wrap(~site, scales = "free_y") + # Compare across sites easily
theme_minimal() +
theme(plot.title = element_text(face = "bold", size = 14))
```

Temporal Variations in Deuterium Excess

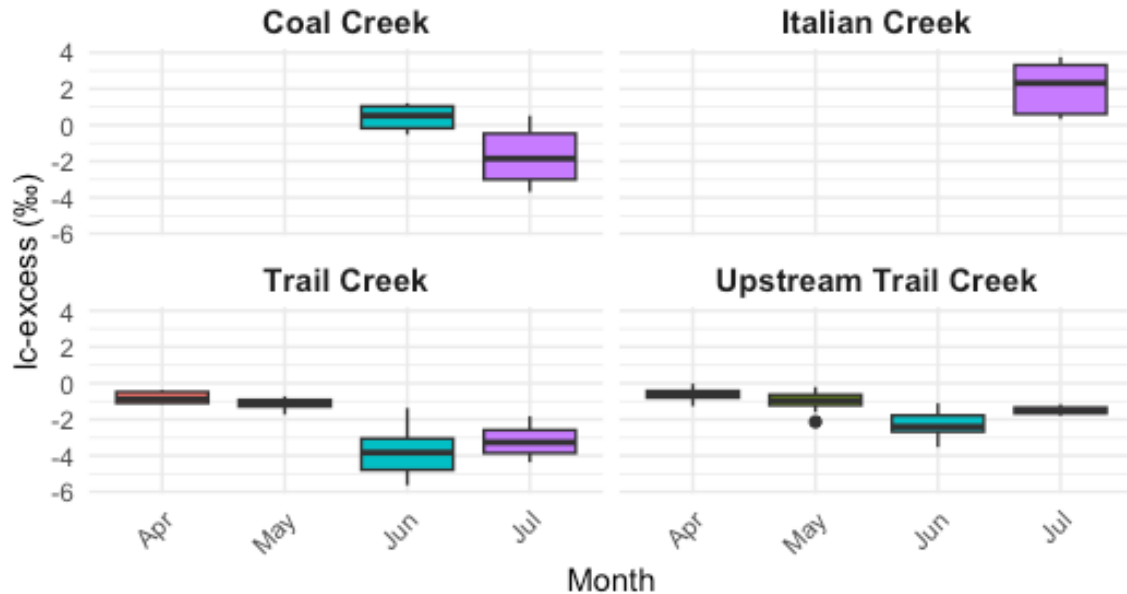


Monthly Variations in Evaporative Fraction (lc-excess) Across Study Sites

```
ggplot(compiled_isotope, aes(x = factor(month(date, label = TRUE, abbr = TRUE)),
y = lc_excess)) +
  geom_boxplot(aes(fill = factor(month(date))), show.legend = FALSE) +
  facet_wrap(~location, labeller = labeller(location = c(tc = "Trail Creek",
cc = "Coal Creek",
it = "Italian Creek",
utc = "Upstream Trail
Creek")))) +
  labs(
    title = "Monthly Variations in Evaporative Fraction (lc-excess) Across
Study Sites",
    subtitle = "Seasonal isotopic shifts indicating changes in local evaporation-
condensation histories",
    x = "Month",
    y = "lc-excess (‰)"
  ) +
  theme_minimal() +
  theme(plot.title = element_text(face = "bold", size = 14),
    strip.text = element_text(face = "bold", size = 11),
    axis.text.x = element_text(angle = 45, hjust = 1)) # tilting the x axis text
```

Monthly Variations in Evaporative Fraction (lc-excess)

Seasonal isotopic shifts indicating changes in local evaporation-condensation



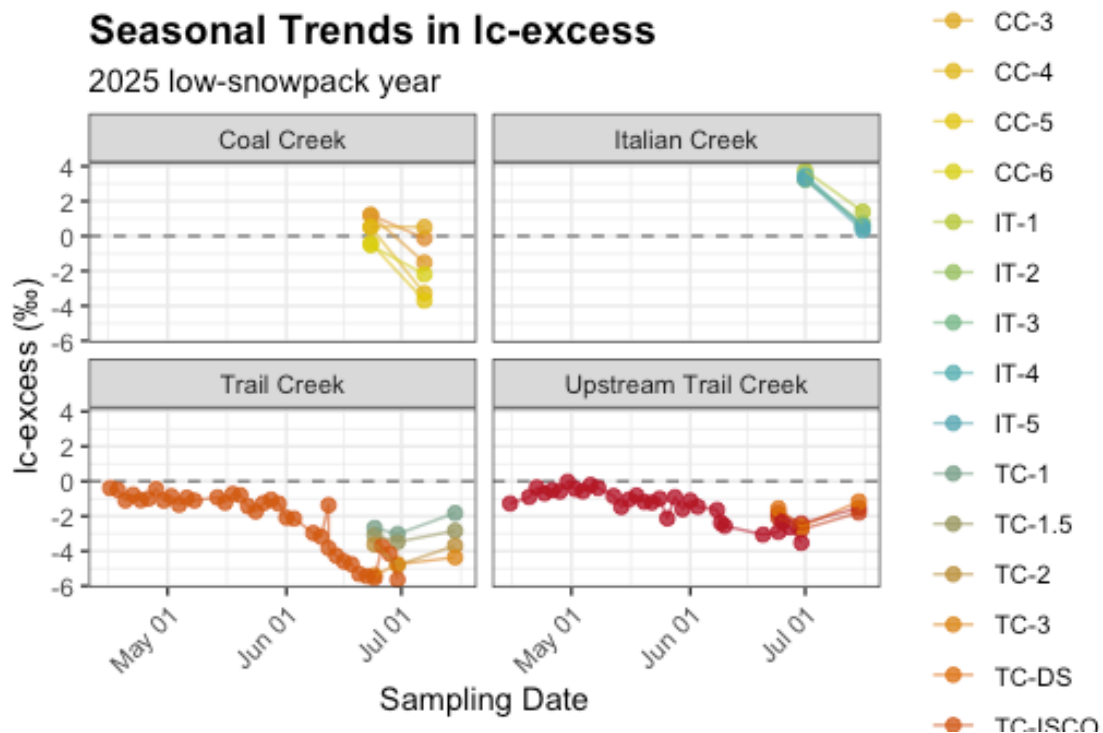
Time series of lc-excess across season, colored by reach, faceted by watershed

```
ggplot(compiled_isotope, aes(x = date, y = lc_excess, color = site, group =
interaction(site, location))) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "gray50") +
  geom_point(size = 2, alpha = 0.8) +
  geom_line(alpha = 0.6) +
  facet_wrap(~location, labeller = labeller(location = c(tc = "Trail Creek",
                                                         cc = "Coal Creek",
                                                         it = "Italian Creek",
                                                         utc = "Upstream Trail
Creek")))) +
  scale_x_date(date_labels = "%b %d") +
  scale_color_manual(values = wes_palette("FantasticFox1", n =
length(unique(compiled_isotope$site)),
                                                         type = "continuous"),
labels = c("coal_1" = "CC-1", "coal_2" = "CC-2", "coal_3"
= "CC-3", "coal_4" =
"CC-4", "coal_5" = "CC-5", "coal_6" = "CC-6",
"italian_1" = "IT-1",
"italian_2" = "IT-2", "italian_3" = "IT-3",
"italian_4" = "IT-4",
"italian_5" = "IT-5", "tc_isco" = "TC-ISCO",
```

```

"tc_1" = "TC-1", "tc_15"
                                = "TC-1.5", "tc_2" = "TC-2", "tc_3" = "TC-3",
"tc_ds" = "TC-DS",
                                "tc_mid" = "TC-Mid", "tc_us" = "TC-US", "utc_isco"
= "UTC-ISCO")) +
  labs(title = "Seasonal Trends in lc-excess",
        subtitle = "2025 low-snowpack year",
        x = "Sampling Date", y = "lc-excess (‰)", color = "Reach") +
  theme_bw() +
  theme(plot.title = element_text(face = "bold", size = 14),
        legend.position = "right",
        axis.text.x = element_text(angle = 45, hjust = 1))

```



Dual-isotope plot ($\delta^{18}\text{O}$ vs δD) with LMWL reference

```

# Add Global Meteoric Water Line (GMWL:  $y = 8x + 10$ )

gmwl_slope <- 8.00
gmwl_intercept <- 10.00

ggplot(compiled_isotope, aes(x = delta_o, y = delta_d, color = site)) +
  geom_point(size = 2.5, alpha = 0.7) +

```

```

geom_abline(slope = gmw1_slope, intercept = gmw1_intercept,
            linetype = "dashed", color = "black", linewidth = 0.8) +
facet_wrap(~location, labeller = labeller(location = c(tc = "Trail Creek",
                                                    cc = "Coal Creek",
                                                    it = "Italian Creek")) +
                                                    utc = "Upstream
Trail Creek",

labs(title = "Dual Isotope Plot:  $\delta^{18}\text{O}$  vs  $\delta\text{D}$ ",
     subtitle = "Dashed line = Global Meteoric Water Line",
     x = expression(paste(delta^18, "O (‰)")),
     y = expression(paste(delta, "D (‰)")),
     color = "Reach") +
theme_bw() +
theme(plot.title = element_text(face = "bold", size = 14),
      legend.position = "none")

```

